

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - Industry based Problem-Solving Question

Course Code: DSA0405 Course Title: FUNDAMENTALS OF DATA SCIENCE

CO Assessed: CO4 – Industry based Problem-Solving Question Assessment: Assessment Tool 2– Industry based Problem-Solving Question

Weightage: 40% of CIA Total Marks: 100

Student Name:	M.Mohamed Bazith	Reg. No.:	192472052
Section / Batch:	CSE-AI	Date:	20-08-2026

Code of Conduct

I M.Mohamed Bazith (192472052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Bazith

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

QUESTIONS :

NOTE: For each problem, find the following — 1. Population and sample 2. Parameter of interest 3. Point estimate 4. Null hypothesis (H_0) 5. Alternative hypothesis (H_1) 6. Significance level (α) 7. Confidence interval and/or test statistic 8. p -value 9. Compare p -value with α 10. Industry-oriented conclusion.

6. SOFTWARE COMPANY – API RESPONSE TIME

Topic: Confidence Interval

A software company measures the response time of 100 API requests. The sample mean is 240 ms and standard deviation is 35 ms. Construct a 95% confidence interval for the average API response time.

7. E-COMMERCE

Topic: Confidence Interval

An e-commerce platform finds that 420 out of 600 randomly selected visitors purchased a product. Construct a 95% confidence interval for the true conversion rate.

8. TELECOM

Topic: Confidence Interval

A telecom company wants to estimate the average monthly data consumption of its users. A sample of 225 users has a mean consumption of 14.8 GB and standard deviation of 4.5 GB. Construct a 99% confidence interval for the population mean.

9. MANUFACTURING

Topic: Hypothesis Testing Using CI

A machine historically produces components with a mean diameter of 50 mm. A sample of 36 components has a mean diameter of 50.8 mm and standard deviation of 2.4 mm. Using a 95% confidence interval, determine whether 50 mm is a plausible population mean.

10. BANKING

Topic: Hypothesis Testing Using CI

A bank claims that its average loan-processing time is 5 days. A sample of 49 applications has an average processing time of 5.6 days with standard deviation 1.4 days. Construct a 95% confidence interval and determine whether the company's claim is reasonable.

ANSWERS:

6. SOFTWARE COMPANY – API RESPONSE TIME (CONFIDENCE INTERVAL)

1. Population and Sample

Population: All API requests processed by the software company's system.

Sample: 100 API requests measured.

2. Parameter of Interest

μ = True mean response time of all API requests.

3. Point Estimate

$\bar{X}$ = 240 ms (sample mean response time)

4–6. Hypotheses and Significance Level

This problem asks only for estimation of the population mean; no specific claimed value is given to test against. Hence, formal hypothesis testing (H_0/H_1 /test statistic/p-value) is not applicable here — only the confidence interval is constructed, using a 95% confidence level ($\alpha = 0.05$).

7. Confidence Interval

$$SE = s / \sqrt{n} = 35 / \sqrt{100} = 35 / 10 = 3.5$$

$$CI = \bar{X} \pm (z \times SE), z = 1.96 \text{ for } 95\% \text{ confidence}$$

$$\text{Margin of Error} = 1.96 \times 3.5 = 6.86$$

$$CI = 240 \pm 6.86 = (233.14, 246.86) \text{ ms}$$

8–9. p-value and Comparison

Not applicable — no claimed value is being tested in this problem.

10. Industry-Oriented Conclusion

The software company can be 95% confident that the true average API response time lies between 233.14 ms and 246.86 ms. This range helps the engineering team benchmark system performance, set realistic Service Level Agreements (SLAs), and identify whether infrastructure upgrades are needed to keep response times within acceptable limits.

7. E-COMMERCE – CONVERSION RATE (CONFIDENCE INTERVAL)

1. Population and Sample

Population: All visitors to the e-commerce platform.

Sample: 600 randomly selected visitors.

2. Parameter of Interest

p = True conversion rate (proportion of visitors who purchase a product).

3. Point Estimate

$\hat{p} = 420 / 600 = 0.70$, i.e., the sample conversion rate is 70%.

4-6. Hypotheses and Significance Level

This problem asks only for estimation of the true conversion rate; no specific claimed value is given to test against. Hence, formal hypothesis testing is not applicable here — only the confidence interval is constructed, using a 95% confidence level ($\alpha = 0.05$).

7. Confidence Interval

$$SE = \sqrt{[\hat{p}(1 - \hat{p}) / n]} = \sqrt{[(0.70 \times 0.30) / 600]} = \sqrt{0.00035} \approx 0.01871$$

$$CI = \hat{p} \pm (z \times SE), z = 1.96 \text{ for } 95\% \text{ confidence}$$

$$\text{Margin of Error} = 1.96 \times 0.01871 \approx 0.0367$$

$$CI = 0.70 \pm 0.0367 = (0.6633, 0.7367), \text{ i.e., } (66.33\%, 73.67\%)$$

8-9. p-value and Comparison

Not applicable — no claimed value is being tested in this problem.

10. Industry-Oriented Conclusion

The e-commerce platform can be 95% confident that the true conversion rate of visitors lies between 66.33% and 73.67%. This range helps the marketing and product teams evaluate the effectiveness of the site design and set realistic targets for future conversion-rate optimization campaigns.

8. TELECOM – MONTHLY DATA CONSUMPTION (CONFIDENCE INTERVAL)

1. Population and Sample

Population: All users of the telecom company.

Sample: 225 randomly selected users.

2. Parameter of Interest

μ = True mean monthly data consumption of all users.

3. Point Estimate

$\bar{X} = 14.8$ GB (sample mean monthly consumption)

4-6. Hypotheses and Significance Level

This problem asks only for estimation of the population mean; no specific claimed value is given to test against. Hence, formal hypothesis testing is not applicable here — only the confidence interval is constructed, using a 99% confidence level ($\alpha = 0.01$).

7. Confidence Interval

$$SE = s / \sqrt{n} = 4.5 / \sqrt{225} = 4.5 / 15 = 0.3$$

$$CI = \bar{X} \pm (z \times SE), z = 2.576 \text{ for } 99\% \text{ confidence}$$

$$\text{Margin of Error} = 2.576 \times 0.3 \approx 0.773$$

$$CI = 14.8 \pm 0.773 = (14.03, 15.57) \text{ GB}$$

8-9. p-value and Comparison

Not applicable — no claimed value is being tested in this problem.

10. Industry-Oriented Conclusion

The telecom company can be 99% confident that the true average monthly data consumption of its users lies between 14.03 GB and 15.57 GB. This estimate supports network capacity planning and helps in designing data plans that match typical customer usage patterns.

9. MANUFACTURING – COMPONENT DIAMETER (HYPOTHESIS TESTING USING CI)

1. Population and Sample

Population: All components produced by the machine.

Sample: 36 randomly selected components.

2. Parameter of Interest

μ = True mean diameter of all components produced by the machine.

3. Point Estimate

$\bar{X}$ = 50.8 mm (sample mean diameter)

4. Null Hypothesis (H_0)

H_0 : $\mu = 50$ mm — the machine's mean diameter has not changed from the historical value.

5. Alternative Hypothesis (H_1)

H_1 : $\mu \neq 50$ mm — the machine's mean diameter has changed from the historical value.

6. Significance Level

$\alpha = 0.05$ (95% confidence)

7. Confidence Interval and Test Statistic

Since the sample size is small ($n = 36$) and the population standard deviation is unknown, a t-distribution is used with $df = 35$.

$$SE = s / \sqrt{n} = 2.4 / \sqrt{36} = 2.4 / 6 = 0.4$$

$$95\% \text{ CI} = \bar{X} \pm (t \times SE), t(0.025, 35) \approx 2.030$$

$$\text{Margin of Error} = 2.030 \times 0.4 \approx 0.812$$

$$\text{CI} = 50.8 \pm 0.812 = (49.99, 51.61) \text{ mm}$$

$$\text{Test Statistic: } t = (\bar{X} - \mu_0) / (s / \sqrt{n}) = (50.8 - 50) / 0.4 = 0.8 / 0.4 = 2.00$$

8. p-value

For a two-tailed test with $t = 2.00$ and $df = 35$: $p\text{-value} \approx 0.053$

9. Compare p-value with α

$0.053 > 0.05$, therefore fail to reject H_0 . This agrees with the confidence interval, since the claimed value of 50 mm lies just within the interval (49.99, 51.61).

10. Industry-Oriented Conclusion

Although the sample mean diameter (50.8 mm) is slightly higher than the historical value of 50 mm, the difference is not statistically significant at the 5% level, since 50 mm remains a plausible value within the confidence interval. The manufacturer does not have sufficient evidence to conclude that the machine's output has shifted, but because the result is close to the significance threshold, continued monitoring of the process is advisable.

10. BANKING – LOAN-PROCESSING TIME (HYPOTHESIS TESTING USING CI)

1. Population and Sample

Population: All loan applications processed by the bank.

Sample: 49 randomly selected loan applications.

2. Parameter of Interest

μ = True mean loan-processing time of all applications.

3. Point Estimate

$\bar{X}$ = 5.6 days (sample mean processing time)

4. Null Hypothesis (H_0)

H_0 : $\mu = 5$ days — the bank's claim about average loan-processing time is true.

5. Alternative Hypothesis (H_1)

H_1 : $\mu \neq 5$ days — the true average loan-processing time differs from the claimed value.

6. Significance Level

$\alpha = 0.05$ (95% confidence)

7. Confidence Interval and Test Statistic

Since the population standard deviation is unknown, a t-distribution is used with $df = 48$.

$$SE = s / \sqrt{n} = 1.4 / \sqrt{49} = 1.4 / 7 = 0.2$$

$$95\% \text{ CI} = \bar{X} \pm (t \times SE), t(0.025, 48) \approx 2.011$$

$$\text{Margin of Error} = 2.011 \times 0.2 \approx 0.402$$

$$\text{CI} = 5.6 \pm 0.402 = (5.198, 6.002) \text{ days}$$

$$\text{Test Statistic: } t = (\bar{X} - \mu_0) / (s / \sqrt{n}) = (5.6 - 5) / 0.2 = 0.6 / 0.2 = 3.00$$

8. p-value

For a two-tailed test with $t = 3.00$ and $df = 48$: $p\text{-value} \approx 0.0043$

9. Compare p-value with α

$0.0043 < 0.05$, therefore reject H_0 . This agrees with the confidence interval, since the claimed value of 5 days lies outside the interval (5.198, 6.002).

10. Industry-Oriented Conclusion

There is sufficient statistical evidence to reject the bank's claim that the average loan-processing time is 5 days. The true average processing time appears to be higher, in the range of approximately 5.2 to 6.0 days. The bank should review its loan-processing workflow and consider process improvements or additional staffing to bring processing times closer to its stated target.